

1. Prove that $f_0 - f_1 + f_2 - \dots - f_{2n-1} + f_{2n} = f_{2n-1} - 1$ where f_i are the Fibonacci numbers
 Fibonacci: $f_0 = 0, f_1 = 1, f_i = f_{i-1} + f_{i-2}$ for $i \geq 2$

Base Case: ($n=1$)

LHS: $f_0 - f_1 + f_2 = 0 - 1 + 1 = 0$

RHS: $f_1 - 1 = 1 - 1 = 0$

$0 = 0$, so base case is true.

Inductive Step:

Assume $f_0 - f_1 + f_2 - f_{2k-1} + f_{2k} = f_{2k-1} - 1$ is true for some $k \geq 1$

Prove that statement is true for $k + 1$, or $f_2(k+1) - 1 - 1 \rightarrow \mathbf{f_{2k+1} - 1}$

$f_0 - f_1 + f_2 - \dots - f_{2k-1} + f_{2k} - f_{2k+1} + f_{2k+2}$

$f_{2k-1} - 1 - f_{2k+1} + f_{2k+2}$ (By induction hypothesis.)

$f_{2k-1} - 1 - (f_{2k} + f_{2k-1}) + (f_{2k+1} + f_{2k})$ (By fibonacci)

$f_{2k-1} - 1 - f_{2k} - f_{2k-1} + f_{2k+1} + f_{2k}$

$- 1 + f_{2k+1}$ or $\mathbf{f_{2k+1} - 1}$

$f_{2k+1} - 1 = f_{2k+1} - 1$

Left hand equals the right hand side, therefore by mathematical induction, the statement is true for all integers larger than one, or $n \geq 1$.

2. Consider the following state machine with six states, labeled 0, 1, 2, 3, 4 and 5. The start state is 0. The transitions are $0 \rightarrow 1, 0 \rightarrow 2, 1 \rightarrow 3, 2 \rightarrow 4, 3 \rightarrow 5, 4 \rightarrow 5$, and $5 \rightarrow 0$. Prove that if we take n steps in the state machine we will end up in state 0 if and only if n is divisible by 4. Argue that to prove the statement above by induction, we first have to strengthen the induction hypothesis. State the strengthened hypothesis and prove it.

Strong Induction:

$P(n)$ and $n \in \mathbb{N}$

$P(1)$ is true and $P(k)$ true for $1 \leq k \leq l, l \in \mathbb{N} \rightarrow P(l+1)$ is true.

Six States: 0, 1, 2, 3, 4, 5

Transitions: $0 \rightarrow 1, 0 \rightarrow 2, 1 \rightarrow 3, 2 \rightarrow 4, 3 \rightarrow 5, 4 \rightarrow 5, 5 \rightarrow 0$

$P(n)$: If n steps from start = 0, will end up at 0 if and only if n is divisible by 4.

- Only states 1 and 2 are accessible, therefore...
- $0 \rightarrow 1$, only 3 is accessible. $n = 1$ and 1 is not divisible by 4 $\rightarrow P(1)$
- $0 \rightarrow 2$, only 4 is accessible. $n = 1$ and 1 is not divisible by 4 $\rightarrow P(1)$
- $1 \rightarrow 3$, only 5 is accessible. $n = 2$ and 2 is not divisible by 4 $\rightarrow P(2)$
- $2 \rightarrow 4$, only 5 is accessible. $n = 2$ and 2 is not divisible by 4 $\rightarrow P(2)$
- $3 \rightarrow 5$, only 0 is accessible. $n = 3$ and 3 is not divisible by 4 $\rightarrow P(3)$
- $4 \rightarrow 5$, only 0 is accessible. $n = 3$ and 3 is not divisible by 4 $\rightarrow P(3)$

Case 1: $4 / k = 0$

By IH, SM is in state 0 after k steps, so it will be in state 1 or state 2 after $k + 1$ steps. Since $4 / k = 0$, follows that $4 / (k+1) = 1$ or 2 .

Case 2: $4 / k = 1$

By IH, SM is in state 1 or state 2 after k steps, so it will be in state 3 or 4 after $k + 1$ steps. Since $4 / k = 1$, follows that $4 / (k+1) = 2$.

Case 3: $4 / k = 2$

By IH, SM is in state 3 or 4 after k steps, so it will be in state 5 after $k + 1$ steps. Since $4 / k = 2$, follows that $4 / k = 4/3$.

Case 4: $4 / k = 4/3$

By IH, SM is in state 5 after k steps, so it will be in state 0 after $k + 1$ steps. Since $4 / k = 4/3$, follows that $4 / (k+1) = 1$, which means it is divisible.

3. Let $P(n)$ be the statement that n can be written as a sum of distinct powers of 2.

For example, $5 = 2^0 + 2^2$ and $10 = 2^1 + 2^3$. Note that $10 = 2^0 + 2^0 + 2^3$ is not a sum of distinct powers of 2. Prove that $P(n)$ is true for all positive integers n by strong induction.

Basis:

($n=1$), $1 = 2^0$ and is a power of 2. Therefore, $P(1)$ is true.

Induction:

For all k such that $1 \leq k \leq n$, $P(k)$ is true and prove $P(k + 1)$.

Case 1: $(k + 1)$ is a power of 2

Statement is true because $k + 1$ itself is a power of 2.

Case 2: $(k + 1)$ is not power of 2.

Find the largest power of 2 less than $k + 1$. Let power be 2^t . $2^t < k + 1$

$k + 1 - 2^t \leq n$, write as the sum of powers of 2.

Let $k + 1 - 2^t = 2^{p_1} + 2^{p_2} + \dots + 2^{p_k}$, as all are powers of 2 and gradually increase.

Add 2^t to both sides, $k + 1 = 2^{p_1} + 2^{p_2} + \dots + 2^{p_k}$

$k = n$, so $n + 1$ is represented by sum of powers of 2.

By Inductive Hypothesis and Mathematical Induction, $P(n)$ is true for all positive integers of n , as every n integer can be written as sums of powers of 2.

4. A robot wanders around a 2-dimensional grid. He starts out at (0,0) and can take the following steps: (+1,+3), (+1,-1), (-4,0) and (0,+4). Define a state machine for this problem. Then, define a Preserved Invariant and prove that the robot will never get to (1,1).

Start State: (0, 0)

Transitions: (x+1, y+3), (x-1, y+1), (x, y-4)

Invariant:

$I = (x + y) \% 2$, representing if the sum of $x + y$ is odd or even,

C1: Move (+1, +3): State = (x + 1, y + 3)

$$(x+1 + y + 3) \% 2 = (x + y) \% 2$$

4 is even, so the invariant remains the same.

C2: Move (+1, -1): State = (x + 1, y - 1)

$$(x+1 + y - 1) \% 2 = (x + y) \% 2$$

Invariant remains the same.

C3: Move (-4, 0): State = (x - 4, y)

$$(x - 4 + y) \% 2 = (x + y) \% 2$$

Invariant remains the same.

C4: Move (0, +4): State = (x, y + 4)

$$(x + y + 4) \% 2 = (x + y) \% 2$$

4 is even, so invariant remains the same.

Initial: Since initial state is (0, 0), the invariant $I = (0 + 0) \% 2 = 0$

Invariant: Remains even for every move as shown above.

Since robot starts with even invariant, will always have even invariant. For point (1, 1), invariant would be $I = (1 + 1) \% 2 = 2 \% 2 = 0$, which is also even. Contradicts requirement for (1, 1) because invariant for (1, 1) is odd.

5. Show that if a predicate $P(n)$ can be proven true for all positive integers n by strong induction, then it can be proven true also by regular induction, once you strengthen the inductive hypothesis. In other words, Strong Induction isn't really stronger than Regular Induction. Hint: Given any statement $P(n)$, define a new (stronger) statement $Q(n)$ so that proving $P(n)$ by strong induction is similar to proving $Q(n)$ by regular induction.

Regular Induction:

Basis: $P(1)$ is true

Inductive Step: $P(n)$ is true and $n = k$, then $P(k + 1)$ is true

Strong Induction:

Basis: $P(1)$ is true

Inductive: $P(n)$ is true for $n \leq k$, then $P(n)$ for $n = k + 1$

If proven by strong induction, then can be proven by regular induction.

Strong Induction: $P(n)$ is true whenever $n \leq k$. So $P(k)$ is true.

If $P(k)$ is true whenever $k \leq n$, then $P(k)$ is true when $n = k$.

$P(k+1)$ is true, so when $n = k$. $P(k)$ is true. Then $P(k+1)$ is true.

Therefore, whenever strong induction is true, regular induction is also true.

So, can prove a predicate by regular induction if proven by strong induction.